Transcendence by Design

Continued Fractions and Diophantine Approximation

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SECTION 1

Continued Fractions

CONTINUED FRACTIONS: Introduction

Given integers $a_0, a_1, \ldots$, a **continued fraction** is an expression of the form

$$[a_0; a_1, \ldots, a_n] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{a_n}}},$$

or

$$\lim_{n \to +\infty} [a_0; a_1, \ldots, a_n] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots}},$$

provided that the limit exists.

The Algorithm

INPUT: α_0

For $n \geq 0$: set $a_n = \lfloor \alpha_n \rfloor$

if $a_n = \alpha_n$: STOP

else: set $\alpha_{n+1} = \frac{1}{\alpha_n - a_n}$

OUTPUT: $a_0, a_1, \ldots$

In practice

$$\alpha_0 = a_0 + \frac{1}{\alpha_1} = a_0 + \frac{1}{a_1 + \frac{1}{\alpha_2}} = \ldots$$

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Example

$$\frac{5}{3} = 1 + \frac{2}{3} = 1 + \frac{1}{\frac{3}{2}} = 1 + \frac{1}{1 + \frac{1}{2}} = [1; 1, 2].$$

Theorem

The continued fraction of α_0 is *finite* if and only if α_0 is *rational*.

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Theorem (Lagrange)

The continued fraction of α_0 is *periodic* if and only if α_0 is *quadratic irrational*.

CONTINUED FRACTIONS: Basic properties

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Theorem (Lagrange)

The continued fraction of α_0 is *periodic* if and only if α_0 is *quadratic irrational*.

Examples

- $\sqrt{2} = [1; \overline{2}]$
- $\frac{1+\sqrt{5}}{2} = [1; \overline{1}]$

CONTINUED FRACTIONS: Convergents

Given $\alpha_0 = [a_0; a_1, a_2, \dots]$, the n-th *convergent* is

$$\frac{p_n}{q_n} := [a_0; a_1, \dots, a_n] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{a_n}}}.$$

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Property

$$\begin{cases} p_n = a_n p_{n-1} + p_{n-2}, \\ q_n = a_n q_{n-1} + q_{n-2}, \end{cases} \qquad \begin{cases} p_{-2} = 0, & p_{-1} = 1 \\ q_{-2} = 1, & q_{-1} = 0 \end{cases} \qquad (n \geq 0).$$

Theorem

$$\left|\alpha_0 - \frac{p_n}{q_n}\right| < \frac{1}{a_{n+1}q_n^2} \leq \frac{1}{q_n^2}, \qquad (n \geq 0)$$

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Theorem (Dirichlet, 1842)

If α_0 is an irrational real number, then there exist *infinitely many* rational numbers p/q such that

$$\left|\alpha_0 - \frac{p}{q}\right| < \frac{1}{q^2}.$$

This is not true if α_0 is rational.

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This is not true if α_0 is rational.

- How much does the quality of the approximation tell about the arithmetic of α_0?

Definition

An element $\alpha_0 \in \mathbb{R}$ is *algebraic* if it is the root of some nonzero polynomial $P(X) \in \mathbb{Q}[X]$. Otherwise, α_0 is transcendental.

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Theorem (Roth, 1955)

Let α_0 be an algebraic number. Then, for every $\varepsilon > 0$, the inequality

$$\left|\alpha_0 - \frac{p}{q}\right| < \frac{1}{q^{2+\varepsilon}}$$

is satisfied **only** for finitely many rational numbers p/q.

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The core idea

Algebraic numbers cannot be approximated “too well” by rational numbers.

TRANSCENDENTAL NUMBERS EXIST

Let $\alpha_0 = [a_0; a_1, a_2, \dots]$ with $\boxed{a_{n+1} > q_n}$. Then, for every $n \geq 0$,

$$\left| \alpha_0 - \frac{p_n}{q_n} \right| < \frac{1}{a_{n+1} q_n^2} < \frac{1}{q_n^3}.$$

By Roth's Theorem, α_0 is transcendental.

Construction with $a_{n+1} = q_n + 1$

n	a_n	q_n
0	1	1
1	2	2
2	3	7
3	8	58
4	59	3429
5	3430	11761528
6	11761529	138333552659741
7	138333552659742	19136171791465474628886608350

A NEW PROBLEM

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Question

Can we find transcendental continued fractions with bounded partial quotients?

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Conjecture (Khintchine, 1964)

The continued fraction of any irrational algebraic number α_0 of degree $d \geq 3$ contains partial quotients of arbitrarily large size.

Definition

Let $(\lambda_k)_{k\geq 0}$, $(r_k)_{k\geq 0}$, $(n_k)_{n\geq 0}$ be sequences of positive integers such that, for all $k \geq 0$,

$$n_{k+1} \geq n_k + r_k\lambda_k.$$

Suppose that

$$a_m = a_{m+r_k} \quad \text{for } n_k \leq m \leq n_k + (\lambda_k - 1)r_k - 1.$$

Then, $[a_0; a_1, a_2, \dots]$ is called *quasi-periodic*.

At position n_k, a word of r_k partial quotients $a_{n_k}, a_{n_k+1}, \ldots, a_{n_k+r_k-1}$ is repeated λ_k times.

$$\alpha = [a_0; \ldots, a_{n_0-1}, \overline{a_{n_0}, \ldots, a_{n_0+r_0-1}}^{\lambda_0}, \ldots, \overline{a_{n_1}, \ldots, a_{n_1+r_1-1}}^{\lambda_1}, \ldots].$$

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Example

$$\left[3; 7, \boxed{1,1}, \boxed{2,2,2}, \boxed{1,1,1,1}, \boxed{2,2,2,2,2}, \dots\right] = [3; 7, \overline{1}^2, \overline{2}^3, \overline{1}^4, \overline{2}^5, \dots].$$

Theorem (Baker, 1962)

Let $\alpha_0 = [a_0; a_1, a_2, \dots]$ be a quasi-periodic (non periodic) continued fraction such that $r_k < c \cdot n_k$ for some constant $c > 0$, and

$$\limsup_{k \to +\infty} \frac{\log \lambda_k \sqrt{\log n_k}}{n_k} = +\infty.$$

Then, α_0 is transcendental.

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Then, α_0 is transcendental.

The core idea

Algebraic numbers cannot be approximated “too well” by algebraic numbers.

SKETCH OF BAKER'S PROOF

Assume α_0 is algebraic of degree d and consider, for every $k \geq 0$,

$$\eta_k := [a_0; a_1, \ldots, a_{n_k-1}, \overline{a_{n_k}, \ldots, a_{n_k+r_k-1}}].$$

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Using the algebraicity of α_0, one obtains

$$|\alpha_0 - \eta_k| \gg \frac{1}{\mathrm{H}(\eta_k)^d} \gg \frac{1}{q_{n_k+r_k-1}^{2d}}.$$

Since $|\alpha_0 - \eta_k| < q_{n_k+r_k\lambda_k-1}^{-2}$, a contradiction is derived.

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Davenport and Roth's Theorem

If α_0 is algebraic, then

$$\log\log q_n \ll \frac{n}{\sqrt{\log n}}$$

SECTION 2

Multidimensional Continued Fractions

The Algorithm

INPUT: α_0, β_0

For $n \geq 0$: set $a_n = \lfloor \alpha_n \rfloor$, $b_n = \lfloor \beta_n \rfloor$

if $b_n \neq \beta_n$: set $\alpha_{n+1} = \dfrac{1}{\beta_n - b_n}$, $\beta_{n+1} = \dfrac{\alpha_n - a_n}{\beta_n - b_n}$

else: compute $\mathrm{CF}(\alpha_n)$

OUTPUT: $\boldsymbol{a} = (a_0, a_1, \dots)$, $\boldsymbol{b} = (b_0, b_1, \dots)$

Example $(\alpha_0, \beta_0) = \left(\sqrt{2}, \frac{1}{3}\right)$

n	a_n	b_n	α_{n+1}	β_{n+1}
0	$\lfloor \sqrt{2} \rfloor = 1$	$\left\lfloor \frac{1}{3} \right\rfloor = 0$	$\frac{1}{\frac{1}{3} - 0} = 3$	$\frac{\sqrt{2} - 1}{\frac{1}{3} - 0} = 3\sqrt{2} - 3$
1	$\lfloor 3 \rfloor = 3$	$\lfloor 3\sqrt{2} - 3 \rfloor = 1$	$\frac{1}{3\sqrt{2} - 4} = \frac{3\sqrt{2} + 4}{2}$	$\frac{3 - 3}{3\sqrt{2} - 4} = \boxed{0}$
2	$\left\lfloor \frac{3\sqrt{2} + 4}{2} \right\rfloor = 4$	$\lfloor 0 \rfloor = 0$	$\frac{1}{\frac{3\sqrt{2}+4}{2} - 4} = 3\sqrt{2} + 4$	–

$$\left(\sqrt{2}, \tfrac{1}{3}\right) = [(1; 3, 4, \overline{8, 4}), (0; 1, 0)]$$

Example $(\alpha_0, \beta_0) = \left(\sqrt{2}, \frac{1}{3}\right)$

$$\sqrt{2} = a_0 + \cfrac{b_1 + \cfrac{1}{a_2 + \cfrac{1}{\alpha_3}}}{a_1 + \cfrac{b_2}{a_2 + \cfrac{1}{\alpha_3}}} = 1 + \cfrac{1 + \cfrac{1}{4 + \cfrac{1}{3\sqrt{2}+4}}}{3 + \cfrac{0}{4 + \cfrac{1}{3\sqrt{2}+4}}} = \dots,$$

$$\frac{1}{3} = b_0 + \cfrac{1}{a_1 + \cfrac{b_2}{a_2 + \cfrac{1}{\alpha_3}}} = 0 + \cfrac{1}{3 + \cfrac{0}{4 + \cfrac{1}{3\sqrt{2}+4}}}.$$

Theorem

Let $(\alpha_0, \beta_0) = [(a_0; a_1, \ldots, a_{k-1}, \overline{a_k, \ldots, a_{k+t}}), (b_0; b_1, \ldots, b_{k-1}, \overline{b_k, \ldots, b_{k+t}})]$.

Then, $[\mathbb{Q}(\alpha_0, \beta_0) : \mathbb{Q}] = 3$.

Theorem

Let $(\alpha_0, \beta_0) = [(a_0; a_1, \ldots, a_{k-1}, \overline{a_k, \ldots, a_{k+t}}), (b_0; b_1, \ldots, b_{k-1}, \overline{b_k, \ldots, b_{k+t}})]$.

Then, $[\mathbb{Q}(\alpha_0, \beta_0) : \mathbb{Q}] = 3$.

The converse is still unknown!

Given $(\alpha_0, \beta_0) = [(a_0; a_1, \dots,), (b_0; b_1, \dots)]$, the n-th convergent is

$$\left(\frac{A_n}{C_n}, \frac{B_n}{C_n}\right) = [(a_0; a_1, \dots, a_n), (b_0; b_1, \dots, b_n)]$$

Given $(\alpha_0, \beta_0) = [(a_0; a_1, \dots,), (b_0; b_1, \dots)]$, the n-th convergent is

$$\left(\frac{A_n}{C_n}, \frac{B_n}{C_n}\right) = [(a_0; a_1, \dots, a_n), (b_0; b_1, \dots, b_n)]$$

Property

$$\begin{cases} A_n = a_n A_{n-1} + b_n A_{n-2} + A_{n-3}, \\ B_n = a_n B_{n-1} + b_n B_{n-2} + B_{n-3} \\ C_n = a_n C_{n-1} + b_n C_{n-2} + C_{n-3} \end{cases} \qquad \begin{cases} A_{-3} = 0, & A_{-2} = 0, & A_{-2} = 1 \\ B_{-3} = 0, & B_{-2} = 1, & B_{-2} = 0 \\ C_{-3} = 1, & A_{-2} = 0, & A_{-2} = 0 \end{cases} \qquad (n \geq 0).$$

MCFs: Approximation properties

For infinitely many $n \in \mathbb{N}$ it holds that

$$\left|\alpha_0 - \frac{A_n}{C_n}\right| < \frac{|A_{n+1}C_n - A_nC_{n+1}|}{a_{n+1}C_n^2} = \frac{|\tilde{A}_{n+1}|}{a_{n+1}C_n^2},$$

$$\left|\beta_0 - \frac{B_n}{C_n}\right| < \frac{|B_{n+1}C_n - B_nC_{n+1}|}{a_{n+1}C_n^2} = \frac{|\tilde{B}_{n+1}|}{a_{n+1}C_n^2}.$$

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Recall: Classical CF

For all $n \in \mathbb{N}$ it holds that

$$\left|\alpha - \frac{p_n}{q_n}\right| < \frac{1}{a_{n+1}q_n^2}.$$

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Recall: Classical CF

For all $n \in \mathbb{N}$ it holds that

$$\left|\alpha - \frac{p_n}{q_n}\right| < \frac{1}{a_{n+1}q_n^2}.$$

Property

$$\begin{cases} \tilde{A}_n = -b_n\tilde{A}_{n-1} - a_{n-1}\tilde{A}_{n-2} + \tilde{A}_{n-3}, \\ \tilde{B}_n = -b_n\tilde{B}_{n-1} - a_{n-1}\tilde{B}_{n-2} + B_{n-3}, \end{cases} \qquad \begin{cases} \tilde{A}_{-3} = 0, \quad \tilde{A}_{-2} = 0, \quad \tilde{A}_{-2} = -1 \\ \tilde{B}_{-3} = 1, \quad \tilde{B}_{-2} = 0, \quad \tilde{B}_{-2} = 0. \end{cases}$$

Theorem (A., Murru and Romeo, 2025)

Let $(\alpha_0, \beta_0) = [(a_0; a_1, \dots), (b_0; b_1, \dots)]$, with $\boxed{a_{n+1} > \max\{|\tilde{A}_{n+1}|C_n^{\varepsilon}, |\tilde{B}_{n+1}|C_n^{\varepsilon}\}}$ for some $\varepsilon > 0$, then α_0, β_0 are $\mathbb{Q}$-linearly independent transcendental numbers.

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Proof. For infinitely many n

$$\left|\alpha_0 - \frac{A_n}{C_n}\right| < \frac{|\tilde{A}_{n+1}|}{a_{n+1}C_n^2} < \frac{1}{C_n^{2+\varepsilon}},$$

$$\left|\beta_0 - \frac{B_n}{C_n}\right| < \frac{|\tilde{B}_{n+1}|}{a_{n+1}C_n^2} < \frac{1}{C_n^{2+\varepsilon}},$$

and the claim follows by Roth's Theorem.

Definition

$[(a_0; a_1, \dots), (b_0; b_1, \dots)]$ is quasi-periodic if $(a_0; a_1, \dots)$ and $(b_0; b_1, \dots)$ are quasi-periodic with the same $(\lambda_k)_{k\geq 0}$, $(r_k)_{k\geq 0}$, $(n_k)_{k\geq 0}$.

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Theorem (A., Murru and Romeo, 2025)

Let $(\alpha_0, \beta_0) = [(a_0; a_1, \dots), (b_0; b_1, \dots)]$ be quasi-periodic (non periodic), such that $a_{n+1} < C_n^{\delta}$ for some $\delta \geq 1$, and $r_k < c \cdot n_k$, where c is a constant. If

$$\lim_{k\to+\infty} \frac{\log \lambda_k}{n_k} = +\infty,$$

then either both α_0 and β_0 are transcendental (and rationally independent), or one of them is cubic irrational and the other is transcendental.

IDEA OF OUR PROOF

Assume at least of α_0 or β_0 is algebraic of degree d and consider

$$\eta_k \coloneqq [a_0; a_1, \ldots, a_{n_k-1}, \overline{a_{n_k}, \ldots, a_{n_k+r_k-1}}];$$
$$\theta_k \coloneqq [b_0; b_1, \ldots, b_{n_k-1}, \overline{b_{n_k}, \ldots, b_{n_k+r_k-1}}]$$

Then, at least one of the following holds

$$\frac{1}{C_{n_k+\lambda_k r_k-1}} \gg |\alpha_0 - \eta_k| \gg \frac{1}{C^{9d}_{n_k+r_k-1}},$$
$$\frac{1}{C_{n_k+\lambda_k r_k-1}} \gg |\beta_0 - \theta_k| \gg \frac{1}{C^{9d}_{n_k+r_k-1}}.$$

Using $\boxed{\log\log C_n \ll n}$, a contradiction is derived.

WISHLIST

Improvement of the results

- Is it still true that $\log\log C_n \ll \dfrac{n}{\sqrt{\log n}}$ for algebraic numbers?
- Is the condition $a_{n+1} < C_n^d$ merely technical?
- What about quasi-periodic *MCFs of higher degree*?

Further promising directions

- Palindromic MCFs?
- Sturmian MCFs?
- Automatic MCFs?

REFERENCES

Accossato, Murru and Romeo (2025). “Transcendence criteria for multidimensional continued fractions”. In: arXiv:2505.03384 - under submission.